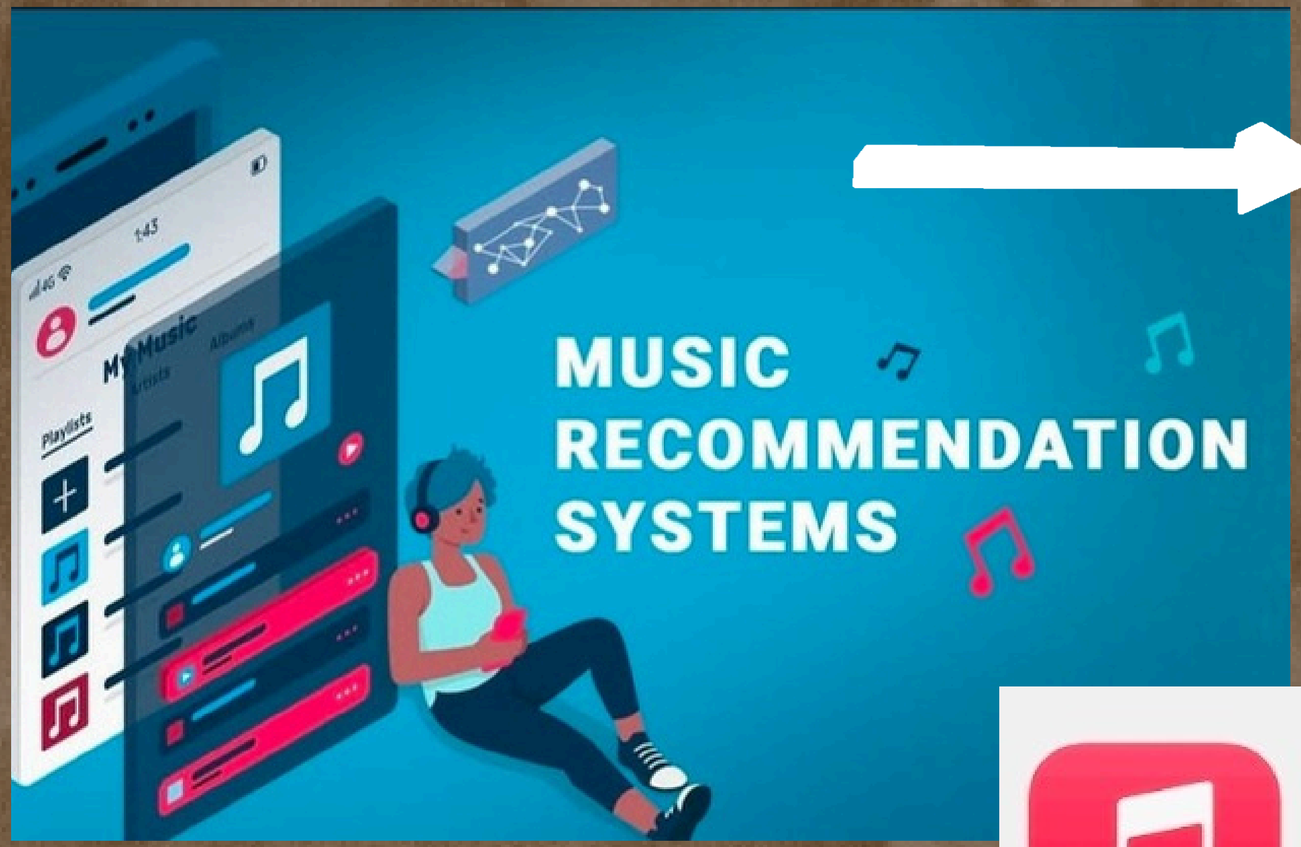


Music Recommendation System

TEAM MEMBERS

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Manandhar Looniva	20253724

Introduction Recap



BASED ON:

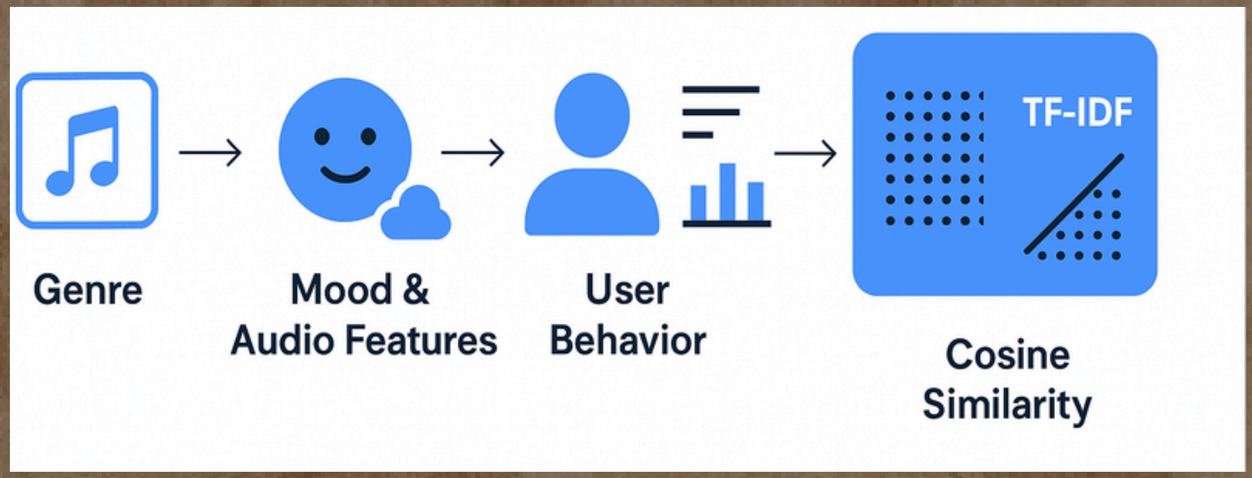
Machine Learning

Listening history

User preferences

Apple Music YouTube Music Spotify

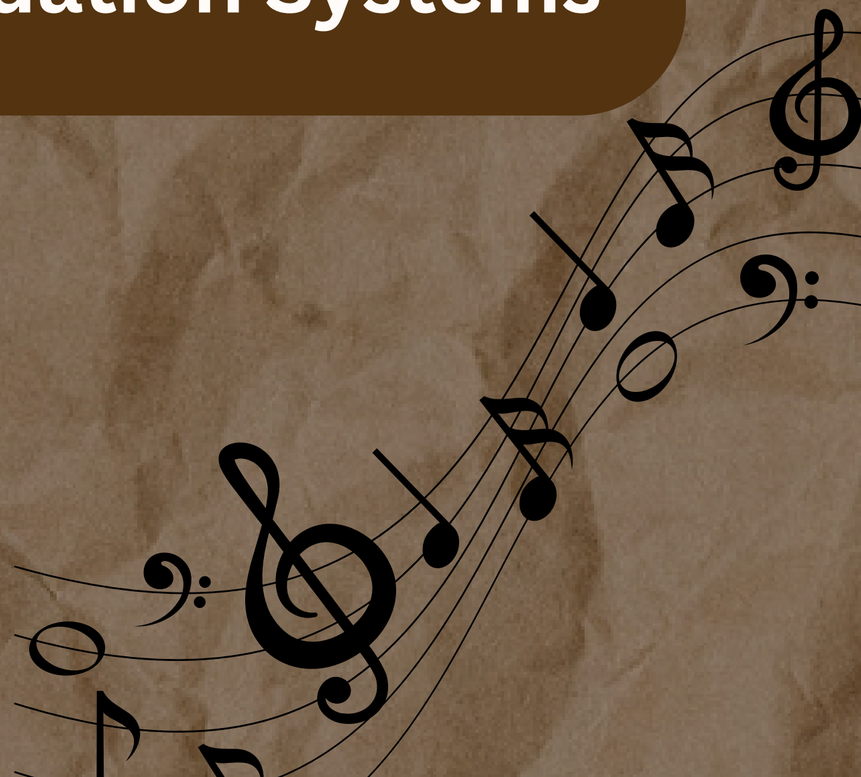
Objectives



- Build a ML-based music recommendation system
- Analyze genre, mood, and user behaviour
- Apply TF-IDF and cosine similarity
- Preprocess and vectorize text data.



“Machine Learning In Music Recommendation Systems”



“Why We Need Music Recommendations”

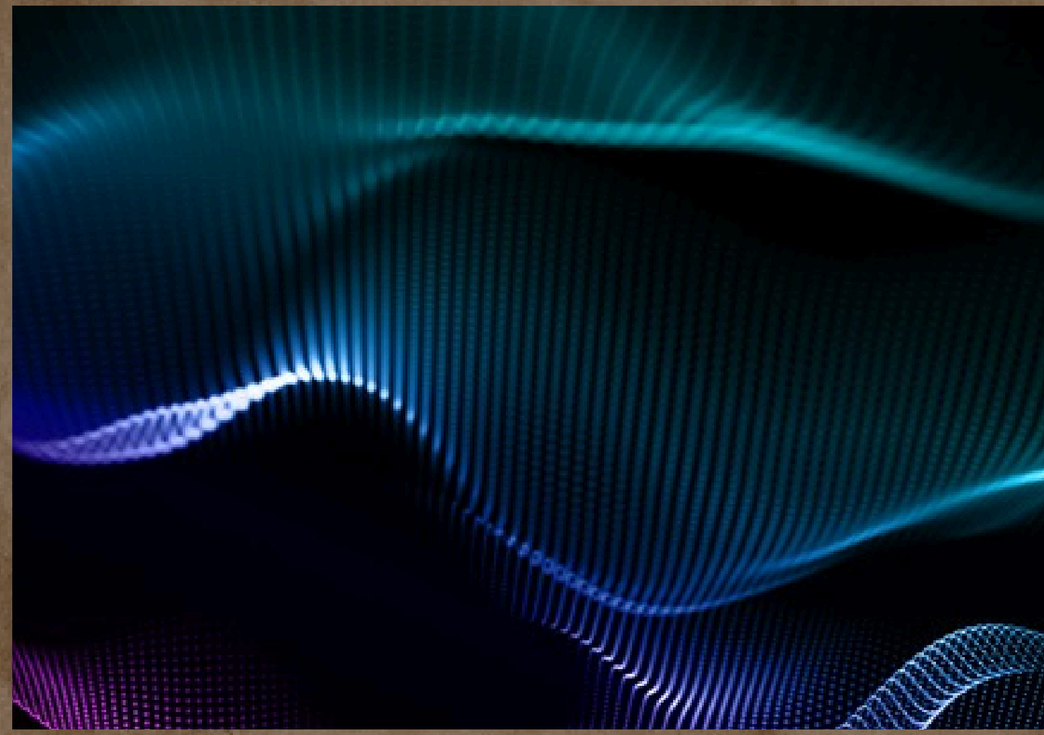
- Millions of songs are released every year
- Hard for users to find music they like on their own
- Machine Learning understands our music taste
- Helps us discover new favorite songs quickly



“How Music Apps Learn Our Taste”

Our listening behavior becomes data :

- “What we listen to often”
- “What we skip immediately”
- “Songs we like or save”
- “Artists and genres we follow”
- “Time and mood of listening”



ML analyzes this to predict what we will enjoy next

“What Machine Learning Learns”

ML learns patterns from our actions :



Play

Songs you finish tell the system what you enjoy

Skips

Skipping fast is strong data - “you don’t like this.”

Likes

Shows strong interest in style, artist, or mood.

Playlist additions

Personal favorites

Then it recommends similar songs

“How Machine Learning Recommends”



1. Collect listening Data
2. Learns User Preference
3. Find similar patterns
4. Recommend Songs

- “Content-Based Filtering: Similar to songs you like.”
- “Collaborative Filtering: What similar users enjoy.”

Dataset Overview

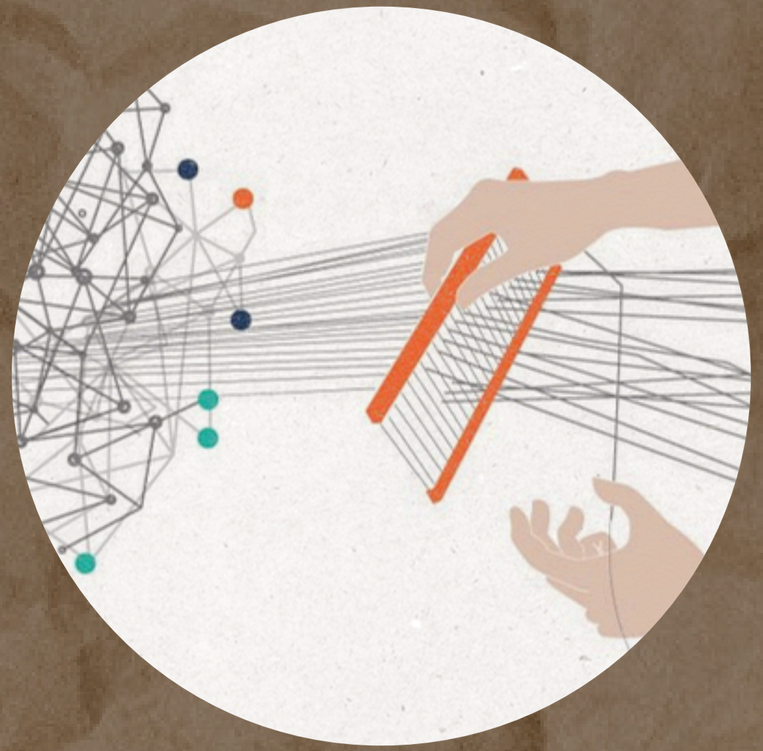
TCC CEDs Music Dataset

- Large music metadata dataset covering ~100 years
- Contains song information such as:
    - Track name
    - Artist name
    - Genre
    - Release details

Combined text-based metadata (track, artist, genre) for similarity analysis, though the dataset is imbalanced with dominant genres/artists.

Text Processing

1. Lowercasing all text for consistency
2. Cleaning special characters
3. Removing missing or invalid metadata



Python Implementation

Core Algorithms

TF-IDF
Cosine similarity

Python Libraries

Pandas
NumPy
Scikit-learn

System Workflow

Preprocess metadata
Apply TF-IDF
Compute similarity matrix
Recommend top similar songs

From Chaos to Clarity: Data Preparation and System Limits



Machine learning

DATA PROCROSSING & CLEANING

Before

Raw data : the starting point



After

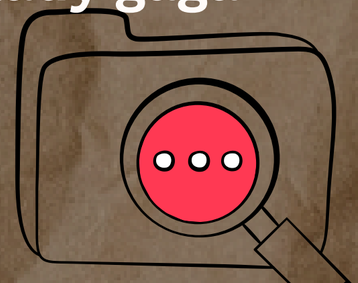
The cleaning procees



- GIGO Rule Applied: Ensuring reliable input.
- De-Duplication: Removing redundant song records.
- Missing Value Handling: Addressing incomplete data.
- Format Standardization: Ensuring consistency.

Lady GaGa & lady gaga

Ready for ML!



Feature Engineering: Combining and Translating Text

Before

Song Title & Artist:
"Cry" - Johnnie Ray

Genre Tags: Blues, Ballad, Pop

Lyrics Snippets:
"Cry... over you... my heart breaks in in two..."



- Rich Input: Music is more than just sound; it's titles, lyrics, and genre tags.
- The Merging Step: Combining all text into a single, comprehensive "document" per song.
- Text Vectorization: Converting this merged text into a numerical format.
- TF-IDF: Assigning importance scores to words for meaningful comparison.

After



Tf Idf vector

0,1 ,0,5 0,8 0.1 ...

Major Challenge: the Cold Start Problem

New user



- Brand new users have not listening histpry
- The system doesn;t know anything about them
- Result : can not personalaize imediately



THE AI IS CLUEES !





- Lack of Interaction Data
- Incomplete Feature Vector
- Difficulty with User Matching

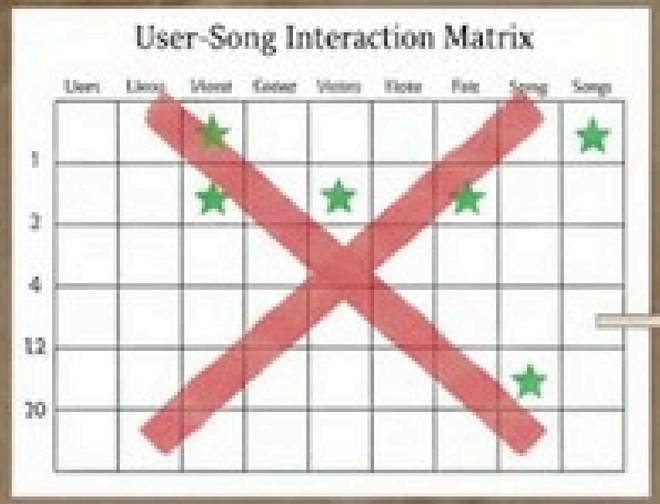
Technical Challenges: Scalabaility & Sparsity

Problem: Immense Data Volume



1 Million Songs + 10 Million Users = TOO MUCH DATA!

Problem: The Sparity Issue



~99% of thi matrix is EMPTY!

- Most users listen to <1% of songs.
- Makes be hard for the AI to find patterns.

CHALLENGE: Finding the needle in the data haystcak!

Model Limitations: The Filter Bubble

Problem: The Filter Bubble



AI traps users in a “bubble” of
of familiar music.

Consequences: Lack of Discovery

- Limits exposure in new genres & artists.
- Reduces sereniguity and surprise.
- Echoes existing tastes, stifling musical growth growth

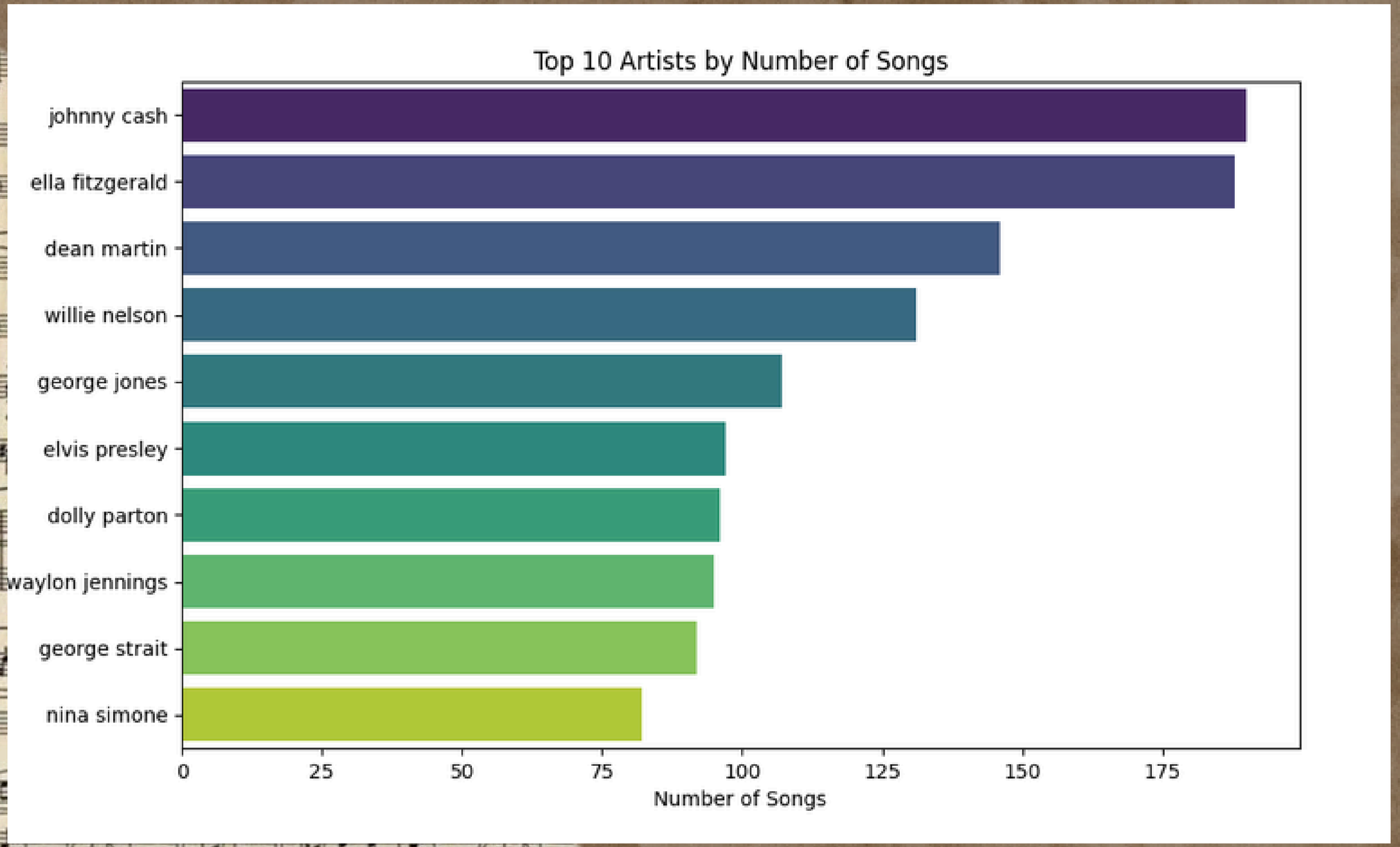
< More the Same

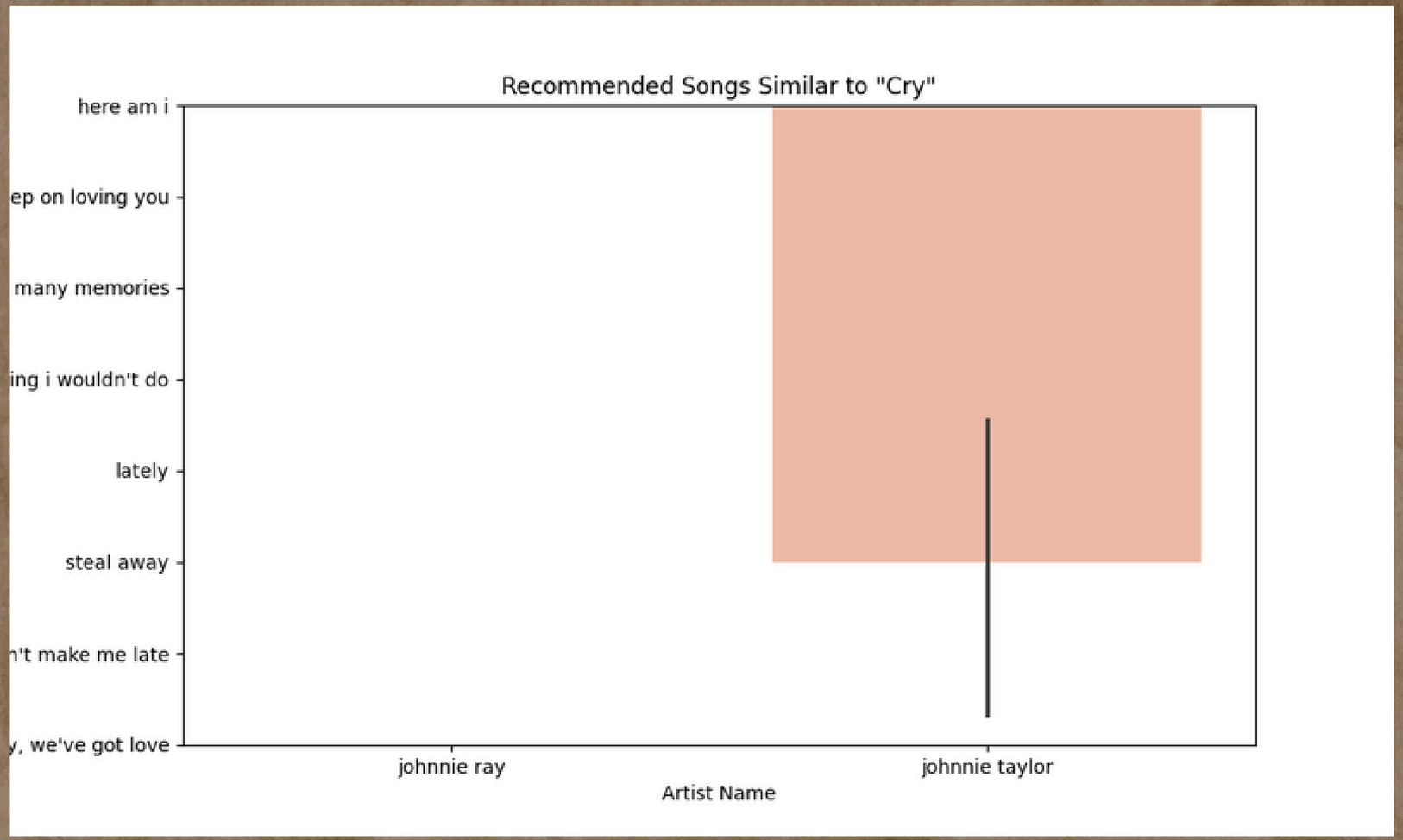
New Music?

The AI struggles to expand your horizons



Result and Discussion


Top 10 Artists by Number of Songs
johnny cash
ella fitzgerald
dean martin
willie nelson
george jones
elvis presley
dolly parton
waylon jennings
george strait
nina simone
0
25
50
75
100
125
150
175
Number of Songs



- Similar genre
- Similar artist style or related artists
- Similar track naming patterns
- Similar text features based on TF-IDF

Characterstics



Similar Songs in Dataset

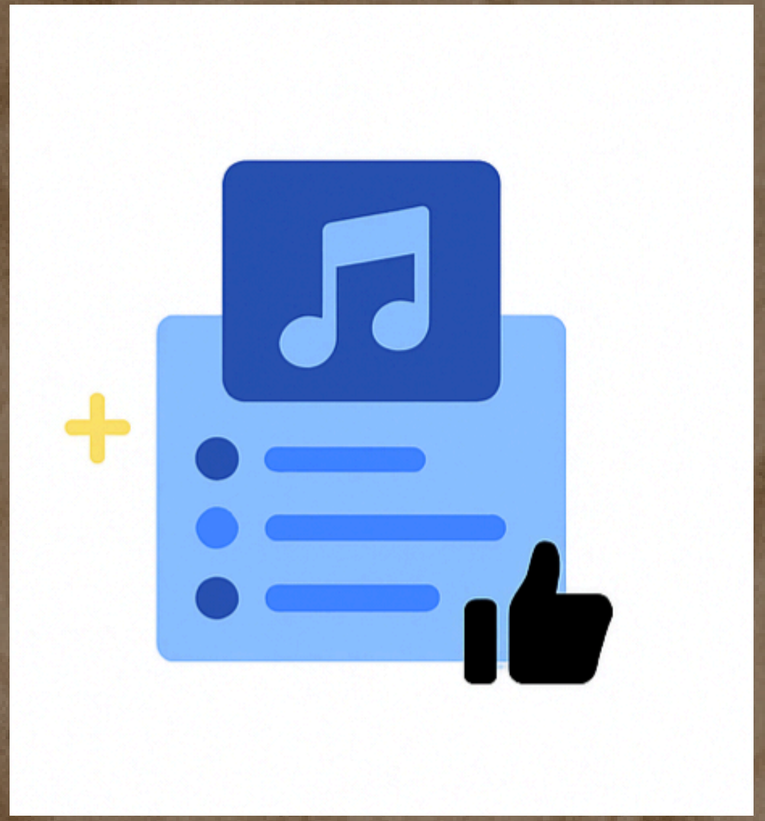
Not based on User preference

Ignores actual audio feature

Conclusion

Findings

• ML technigues like TF-IDF and cosine similarity effectively identified relationships between songs
• Combining genre, mood/audio features, and user behavior produced more accurate and personaliz-ed recommendations



Importance of Recommendation Systems

• Enhance user experience through personalization
• Reduce search time and improve platform engagement
• Widely used in entertainment, e-commerce, and social platforms

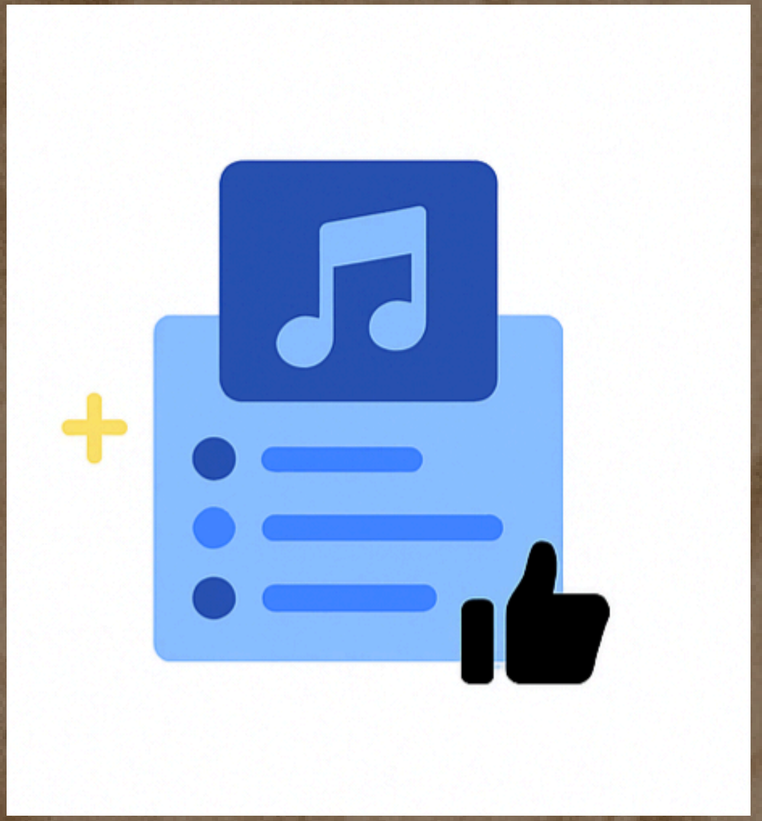
Conclusion

What the Project Achieved:

Designed a complete workflow from data preprocessing → Vectorization → similarity matching

Implemented the system using Python (Pandas, NumPy, Scikit-learn)

Produced accurate example recommendations



Thank You

